

Задание 1

```
#include <iostream>
using namespace std;

int main()
{
    int x;
    cin >> x;
    if (x >= 0)
    {
        cout << "Positive";
    }
    else
    {
        cout << "Negative";
    }
}
```

Задания if

1)

```
#include <iostream>
using namespace std;

int main()
{
    int x;
    cin >> x;
    if (x > 0)
    {
        cout << "Positive";
    }
    else if (x == 0)
    {
        cout << "= 0";
    }
    else
    {
        cout << "Negative";
    }
}
```

```

#include <iostream>
using namespace std;

int main()
{
    int x;
    cin >> x;
    if (x > 0)
    {
        cout << "Positive";
    }
    else if (x == 0)
    {
        cout << "= 0";
    }
    else
    {
        cout << "Negative";
    }
}

```

2)

```

#include <iostream>
using namespace std;

```

```

int main()
{
    int x;
    cin >> x;
    if (x % 2 == 0)
    {
        cout << "Chetnoe";
    }
    else
    {
        cout << "Nechetnoe";
    }
}

```

```

#include <iostream>
using namespace std;

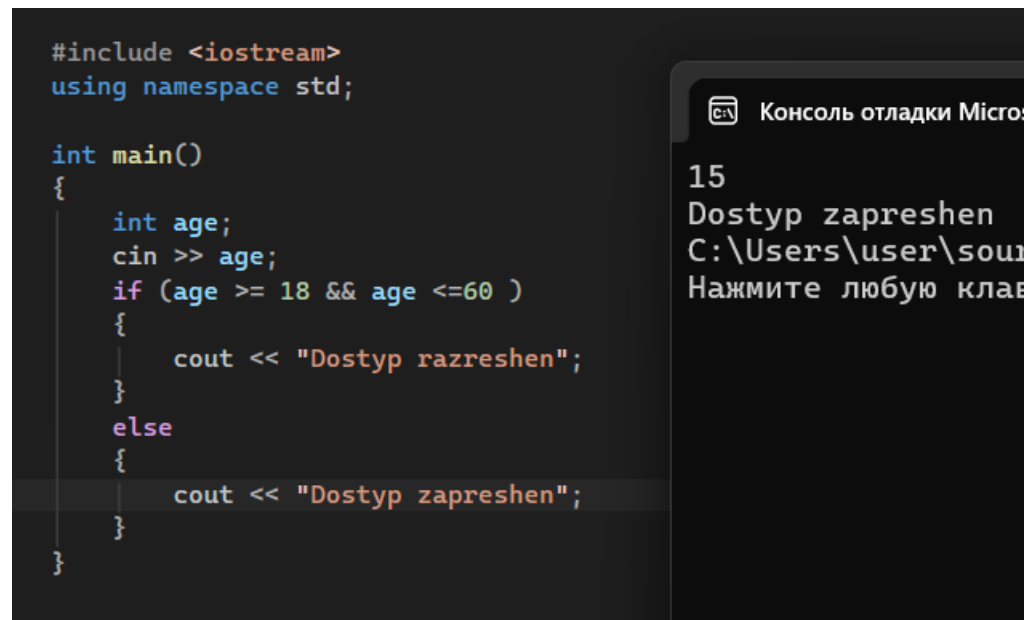
int main()
{
    int x;
    cin >> x;
    if (x % 2 == 0)
    {
        cout << "Chetnoe";
    }
    else
    {
        cout << "Nechetnoe";
    }
}

```

3)

```
#include <iostream>
using namespace std;
```

```
int main()
{
    int age;
    cin >> age;
    if (age >= 18 && age <=60 )
    {
        cout << "Dostyp razreshen";
    }
    else
    {
        cout << "Dostyp zapreshen";
    }
}
```

The image shows a screenshot of a C++ program being executed in a debugger. On the left, the source code is displayed with syntax highlighting. The program includes <iostream>, uses the std namespace, and has a main function that reads an integer 'age' from the user. It then checks if 'age' is between 18 and 60. If true, it prints "Dostyp razreshen"; otherwise, it prints "Dostyp zapreshen". On the right, the "Консоль отладки Microsoft" (Microsoft Debug Console) is open, showing the program's output. The first line is "15", which is the input value for 'age'. The second line is "Dostyp zapreshen", which is the output of the program since 15 is less than 18. Below the output, the file path "C:\Users\user\source\code\" and a prompt "Нажмите любую клавишу" (Press any key) are visible.

```
#include <iostream>
using namespace std;

int main()
{
    int age;
    cin >> age;
    if (age >= 18 && age <=60 )
    {
        cout << "Dostyp razreshen";
    }
    else
    {
        cout << "Dostyp zapreshen";
    }
}
```

Консоль отладки Microsoft

15
Dostyp zapreshen
C:\Users\user\source\code\
Нажмите любую клавишу

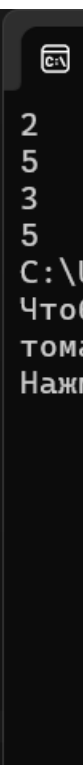
4)

```
#include <iostream>
using namespace std;

int main()
{
    int a, b, c, max;
    cin >> a >> b >> c;
    if (b > a && c < b)
    {
        max = b;
    }
    else if (c > b && c > a)
    {
        max = c;
    }
    else
    {
        max = a;
    }
    cout << max;
}
```

```
#include <iostream>
using namespace std;

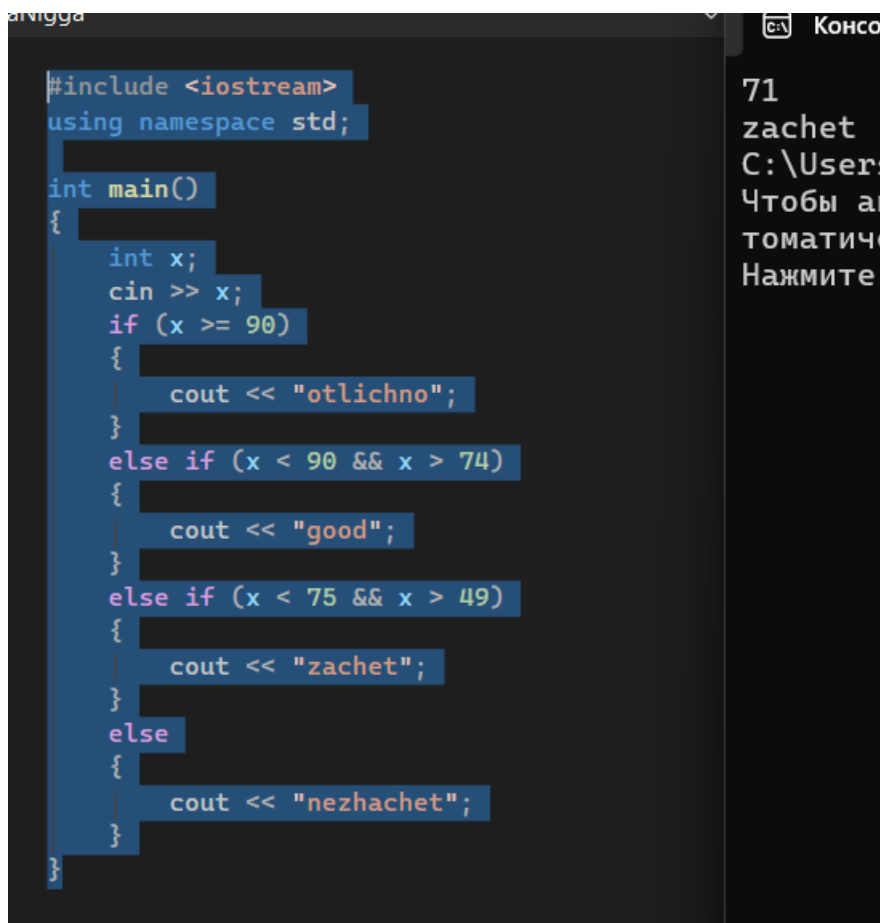
int main()
{
    int a, b, c, max;
    cin >> a >> b >> c;
    if (b > a && c < b)
    {
        max = b;
    }
    else if (c > b && c > a)
    {
        max = c;
    }
    else
    {
        max = a;
    }
    cout << max;
}
```



5)

```
#include <iostream>
using namespace std;

int main()
{
    int x;
    cin >> x;
    if (x >= 90)
    {
        cout << "otlichno";
    }
    else if (x < 90 && x > 74)
    {
        cout << "good";
    }
    else if (x < 75 && x > 49)
    {
        cout << "zachet";
    }
    else
    {
        cout << "nezhachet";
    }
}
```



The screenshot shows a code editor with a dark background. The code is the same as in the previous block. On the right side, there is a console window titled "Консоль" (Console). It displays the output of the program: "71 zachet". Below this, there is some text in Russian: "C:\Users...", "Чтобы а...", "томатич...", "Нажмите...".

Выражение	Результат (true/false)
age >= 18	true
age >= 18 && age <= 60	true
score >= 50 age >= 21	true
!blocked	true
age < 18 score >= 70	true

Исправленная программа

```
#include <iostream>
using namespace std;

int main()
{
    int age;
    cin >> age;

    if (age == 18)
    {
        cout << "Возраст равен 18";
    }
    else
    {
        cout << "Возраст не равен 18";
    }

    return 0;
}
```

Оператор = присваивает значение, а == проверяет. Ошибка встречается часто либо из-за невнимательности, либо из-за того что люди воспринимают = как равно

Задания switch

```
1)
#include <iostream>
using namespace std;

int main()
{
    int x = 1;
    cin << x;
    switch (x)
    {
        case 1:
            cout << "Start";
            break;
        case 2:
            cout << "Settings";
            break;
        case 3:
            cout << "Exit";
            break;
        default:
            cout << "Unknown command";
    }
}
```

```
#include <iostream>
using namespace std;

int main()
{
    int x = 1;
    cin >> x;
    switch (x)
    {
        case 1:
            cout << "Start";
            break;
        case 2:
            cout << "Settings";
            break;
        case 3:
            cout << "Exit";
            break;
        default:
            cout << "Unknown com
    }
}
```

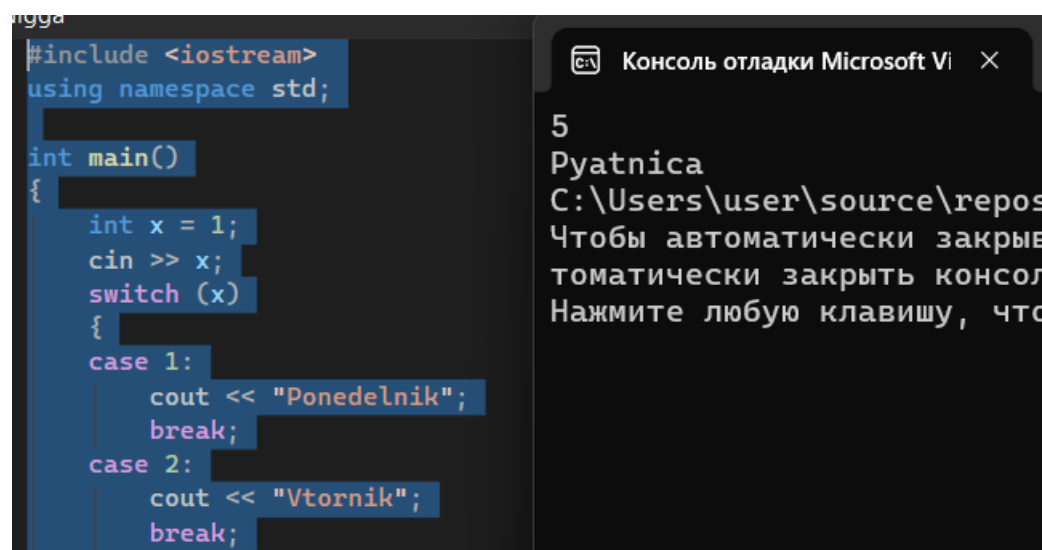
Код

3
Exit
C:\Use
Чтобы
томат
Нажми

2)

```
#include <iostream>
using namespace std;

int main()
{
    int x = 1;
    cin >> x;
    switch (x)
    {
        case 1:
            cout << "Ponedel'nik";
            break;
        case 2:
            cout << "Vtornik";
            break;
        case 3:
            cout << "Sreda";
            break;
        case 4:
            cout << "Chetverg";
            break;
        case 5:
            cout << "Pyatnica";
            break;
        case 6:
            cout << "Subbota";
            break;
        case 7:
            cout << "Voskresenie";
            break;
    }
}
```



```
#include <iostream>
using namespace std;

int main()
{
    int x = 1;
    cin >> x;
    switch (x)
    {
        case 1:
            cout << "Ponedel'nik";
            break;
        case 2:
            cout << "Vtornik";
            break;
```

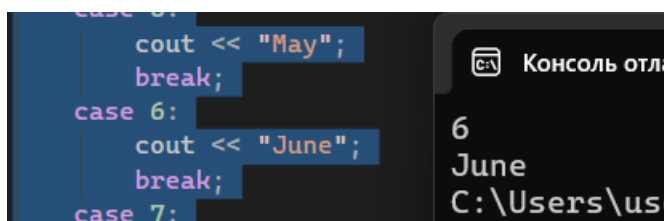
Консоль отладки Microsoft Visual Studio

```
5
Pyatnica
C:\Users\user\source\repos\...
Чтобы автоматически закрыть консоль, нажмите любую клавишу, что...
```

3)

```
#include <iostream>
using namespace std;

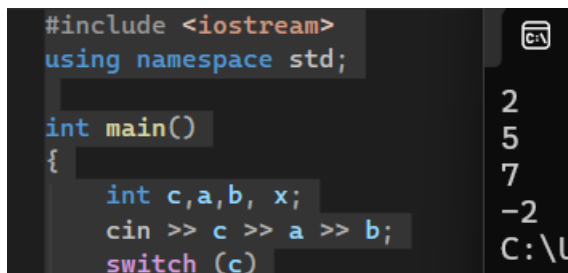
int main()
{
    int x = 1;
    cin >> x;
    switch (x)
    {
        case 1:
            cout << "January";
            break;
        case 2:
            cout << "February";
            break;
        case 3:
            cout << "March";
            break;
        case 4:
            cout << "Aprel";
            break;
        case 5:
            cout << "May";
            break;
        case 6:
            cout << "June";
            break;
        case 7:
            cout << "July";
            break;
        case 8:
            cout << "August";
            break;
        case 9:
            cout << "September";
            break;
        case 10:
            cout << "October";
            break;
        case 11:
            cout << "November";
            break;
        case 12:
            cout << "December";
            break;
    }
}
```



4)

```
#include <iostream>
using namespace std;
```

```
int main()
{
    int c,a,b, x;
    cin >> c >> a >> b;
    switch (c)
    {
        case 1:
            x = a + b;
            break;
        case 2:
            x = a - b;
            break;
        case 3:
            x = a * b;
            break;
        case 4:
            x = a / b;
            break;
    }
    cout << x;
}
```



```
#include <iostream>
using namespace std;

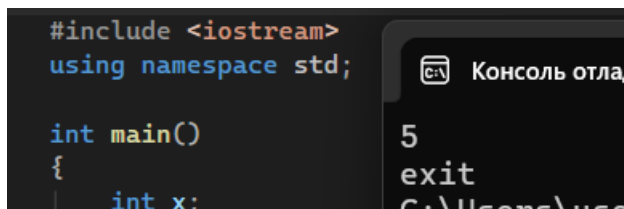
int main()
{
    int c,a,b, x;
    cin >> c >> a >> b;
    switch (c)
```

2
5
7
-2
C:\u

5)

```
#include <iostream>
using namespace std;
```

```
int main()
{
    int x;
    cin >> x;
    switch (x)
    {
        case 1:
            cout << "profile";
            break;
        case 2:
            cout << "messages";
            break;
        case 3:
            cout << "settings";
            break;
        case 4:
            cout << "help";
            break;
        case 5:
            cout << "exit";
            break;
        default:
            cout << "error";
    }
}
```



The screenshot shows a code editor with the following C++ code:

```
#include <iostream>
using namespace std;

int main()
{
    int x;
```

To the right of the code editor is a console window titled "Консоль отла" (Debug Console). It displays the output of the program:

```
5
exit
C:\Users\user
```

Ситуация	if / switch
Проверить age >= 18	if
Выбрать пункт меню 1–5	switch
Проверить score >= 50 && score <= 100	if
Вывести день недели по номеру	switch
Проверить пароль и логин одновременно	if
Выбрать месяц 1–12	switch

Switch лучше подходит если надо проверить конкретные одиночные значения а if для объёмных проверок